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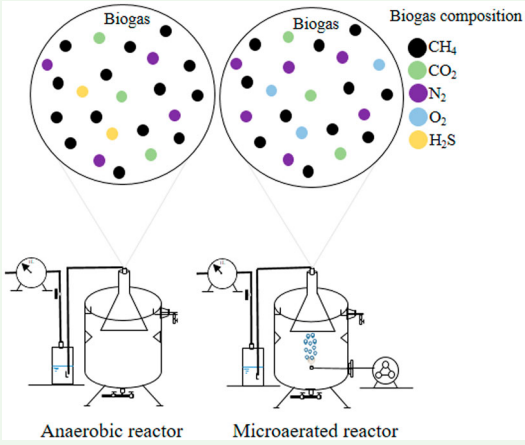
Evaluation of microaeration strategies in the digestion zone of UASB reactors as an alternative for biogas desulfurization

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ABSTRACT

This study aimed at evaluating the microaeration as an alternative for hydrogen sulfide removal from biogas of UASB reactors treating sewage. The set-up consisted of two pilot-scale UASB reactors, including a conventional anaerobic and a modified UASB reactor, operated under microaerated conditions. Air was supplied in the digestion zone, at 1 and 3 m from the bottom of the reactor, and three different air flows were investigated: 10, 20, and 30 $mL.min^{-1}$, corresponding to 0.003, 0.005 and 0.005 $LO_2/L_{influent}$, respectively. The main results showed that the microaeration provided a substantial decrease in hydrogen sulfide concentrations when compared to the concentrations observed in the biogas of the anaerobic UASB reactor. Hydrogen sulfide concentrations remained below 70 ppm_v throughout the experimental period, corresponding to an average removal efficiency of 98%. Although a decrease in methane concentrations in biogas was observed, the feasibility of energy use would not be affected. The effect of microaeration on the overall performance of the reactor was evaluated, however, no significant differences were observed. The feasibility of limiting aeration conditions in the reactor digestion zone as an efficient alternative for hydrogen sulfide removal from biogas was demonstrated.



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UASB reactors; biogas; hydrogen sulfide; microaeration; energy use

Introduction

The process of anaerobic digestion may be adopted to treat solids or liquids. It is possible to assume that this process is an alternative for resources and energy recovery, for instance, converting organic carbon into methane, the main constituent of biogas. Biogas production makes anaerobic digestion an attractive technology to bioenergy generation and a promising alternative to fossil fuels replacement [1–3]. Due to its high methane content, the biogas may be used for electrical and thermal energy generation in cogeneration plants or may be injected into the natural gas network. For both applications, the removal of undesirable compounds, such as water vapor, carbon dioxide, and especially hydrogen sulfide, is required [4].

Sulfur compounds present in sewage are not desired in anaerobic treatment processes, notably in by-products such as treated effluent and biogas. Oxidized sulfur compounds (i.e sulfates) are biochemically reduced to sulfide under anaerobic conditions and further transferred to biogas in the form of gaseous hydrogen sulfide (H_2S) [5,6]. Therefore, the biogas

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produced by the anaerobic digestion of domestic sewage is typically composed of methane (70-80%), nitrogen (10-25%), and carbon dioxide (5-10%) [7]. It is noteworthy that trace amounts of hydrogen sulfide (H_2S) in biogas may affect energy conversions when present in concentrations greater than 300-500 ppm_v, depending on the equipment. Due to its corrosive characteristics, low sulfide concentrations may result in the deterioration of boilers, pipes, and motors [8]. Therefore, methods that provide the selective removal of this compound aiming at biogas enrichment are required [6].

Sulfide biological oxidation is a low-cost clean alternative that promotes sulfide removal from liquid and/or gaseous phases along with the possibility of sulfur recovery [9]. On the other hand, processes involving either chemical or physicochemical oxidation of sulfide demand high energy and require chemical products, possibly leading to the formation of toxic by-products [10–12]. The adopted pathway to remove the sulfide produced in anaerobic reactors should search for low-cost and sustainable technologies, thus requiring simple and robust alternatives, [12] notably in cases of technologies that may be applied in developing countries, such as Brazil.

Previous studies [13–17] have demonstrated that microaeration is a promising technique regarding the oxidation of sulfide produced in the anaerobic treatment of sulfate-containing wastewaters. The authors observed that air/oxygen introduction in the liquid phase not only provided the oxidation of gaseous hydrogen sulfide biogas but also of dissolved sulfide present in the liquid phase. As Equation 1 shows, limiting oxygen concentrations favor sulfide oxidation to elemental sulfur, an insoluble and recoverable compound.

$$2HS^- + O_2 = 2S^0 + 2OH^- \quad (1)$$

According to Krayzelova et al. [18], the range of air/oxygen adopted is crucial in microaerated systems. The authors established the microaeration range from the ratio of oxygen supplied and influent volume, resulting in the range of 0.03 to 1.27 $LO_2/L_{influent}$. The range determined by Krayzelova et al. [18] have been investigated in anaerobic reactors fed on sulfate-rich wastewaters (750–4,500 $mgSO_4^{2-}.L^{-1}$) [15,17,19–21]. Lower ratios (0.0098–0.001 $LO_2/L_{influent}$) have been investigated in anaerobic reactors fed on brewery [22] and synthetic wastewaters [23], with sulfate concentrations ranging from 84 and 143 $mgSO_4^{2-}.L^{-1}$, respectively.

The microaeration has been widely applied in anaerobic sludge digesters, however, studies investigating the performance of microaeration in anaerobic reactors treating domestic sewage with low sulfate concentrations are scarce [23–25]. Therefore, further studies aiming at assessing the effect of microaeration in UASB reactors, regarding hydrogen sulfide oxidation from both liquid and gaseous phases are needed and of great relevance. This technique would not only improve the feasibility of electrical and thermal energy generation but would also provide elemental sulfur recovery.

The main goal of this study was to evaluate different microaeration strategies for hydrogen sulfide removal from biogas, by adding air in the digestion zone of a UASB reactor treating sewage. The technical feasibility of biogas use under the tested conditions was assessed as well as the overall performance of the reactor in terms of organic matter removal and sulfur conversions.

Material and methods

Experimental set-up

The set-up was installed at the Center for Research and Training in Sanitation - UFMG/COPASA, located at the Arrudas sewage treatment plant (Belo Horizonte, Brazil). The biomass of the UASB reactors was due to the retention of microorganisms naturally present in raw sewage once the reactors were not inoculated. The pilot-scale reactors have been continuously operated for 2 years. The UASB reactors were fed with an aliquot of the domestic sewage taken after being submitted to preliminary treatment for solids and grit removal. The main characteristics of the typical urban sewage were: pH – 7.6; COD – 515.7 $mg.L^{-1}$; S^{2-} – 0.5 $mg.L^{-1}$; SO_4^{2-} – 32 $mg.L^{-1}$, as shown in Table 1.

The set-up of this study was comprised of two UASB reactors, as shown in Figure 1. One of the reactors was modified to operate under microaerated conditions. The microaeration system consisted of a peristaltic pump (Masterflex® console drive 7518-10), connected by tubing (Masterflex® 6402-13 L/S tubing, Norprene®) to an air diffuser (ceramic porous stone - 3.5 cm/1.38'' x 2 cm/0.79'') installed inside the reactor. Dimensional

Table 1. Composition of raw sewage influent to the UASB reactors.

Parameters	Average	Median	Standard deviation	Maximum	Minimum
pH	7.6	7.6	0.2	8.0	7.2
Total COD ($mgCOD.L^{-1}$)	543.0	515.7	181.5	895.7	309.0
Sulfide ($mg\ S^{2-}.L^{-1}$)	0.6	0.5	0.5	1.8	0.0
Sulfate ($mg\ SO_4^{2-}.L^{-1}$)	32.3	32.0	4.2	46.5	21.8

Note: Values obtained from samples collected between 9 and 11 a.m during the monitored period.

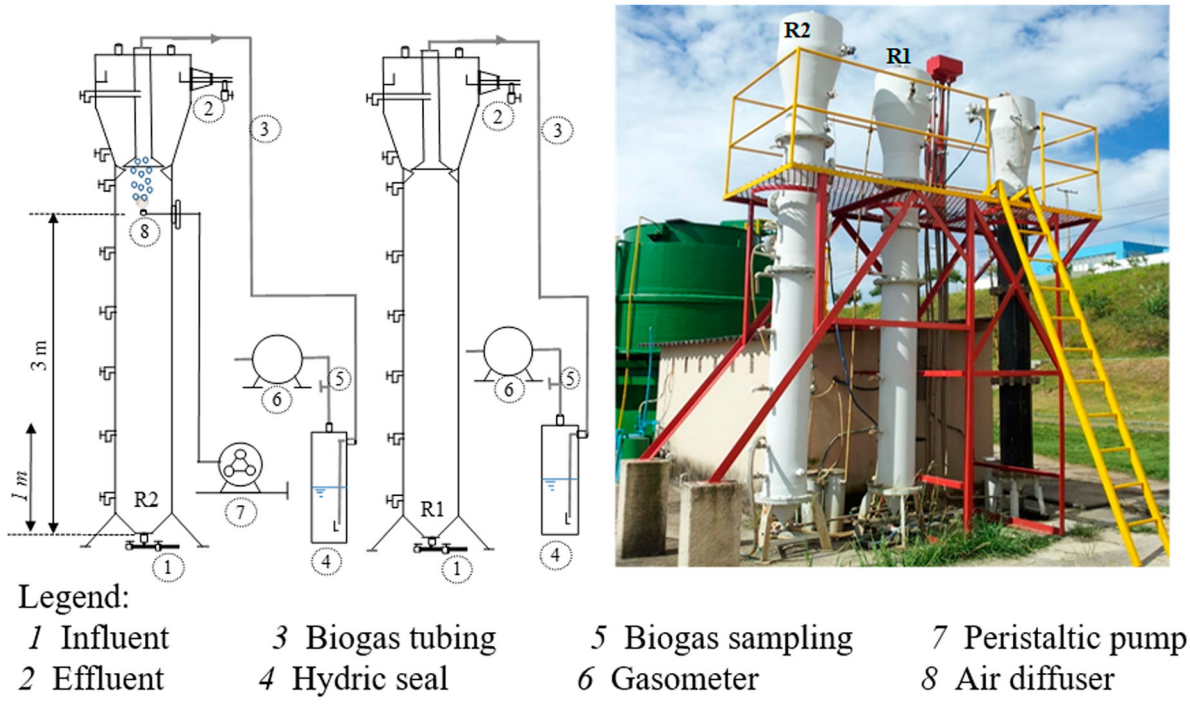


Figure 1. Schematic and real representations of the UASB reactors: R1-anaerobic and R2-microaerated.

and constructive aspects of the investigated UASB reactors are presented in Table 2.

The airflow rate added by influent sewage flow (LO_2/L_{sewage}) was established from the molar ratio between oxygen and sulfur, assuming that all sulfate present in the influent would firstly be reduced to sulfide and further oxidized to elemental sulfur, according to the stoichiometric equations (2) and (3).

$$SO_4^{-2} + \text{Organicmatter} \xrightarrow{\text{Anaerobic bacteria}} HS^- + HCO_3^- \quad (2)$$

$$2HS^- + O_2 \rightarrow 2S^0 + 2OH^- \quad (3)$$

The sulfide produced under anaerobic conditions via sulfate reduction was estimated according to influent sulfate concentration (see Table 3). Therefore, 1 mol of influent sulfate corresponded to 1 mol of sulfide produced. Considering 800 mL.min^{-1} of sewage and 50 mgSO_4^{2-}.L^{-1}, 0.0004 mol sulfide is produced per minute. The molar volume of 22.4 L.mol^{-1} of gas in STP leads to an oxygen flow of 4.5 mL.min^{-1}, corresponding to the airflow of 21.4 mL.min^{-1}. Airflows adopted in the following operational phases are presented in Table 3.

The microaeration was applied at two different heights of the digestion zone of the reactor: at 3 m from the bottom of the reactor, in the sludge blanket, and at 1 m from the bottom of the reactor, in the sludge bed. The tested operational conditions are summarized in Table 3.

Table 2. Dimensional and constructive characteristics of the anaerobic (R1-anaerobic) and microaerated (R2-microaerated) UASB reactors.

	Pilot-scale reactors	
Characteristics	R1-anaerobic	R2-microaerated
Diameter (m)	0.30	0.30
Useful height (m)	4.0	4.25
Useful volume (L)	340	364
Material	Polyethylene/fiberglass	Fiberglass
Air supply	Absent	Present

Analytical methods

The biogas composition in terms of hydrogen sulfide (H_2S), methane (CH_4), nitrogen (N_2), oxygen (O_2), and

Table 3. Operational conditions applied to the microaerated UASB reactor (R2-microaerated) throughout the experimental period.

Phases	P1	P2	P3	P4
HRT (h)	7	7	5	5
Sewage (mL.min^{-1})	800	800	1,200	1,200
Raw sewage (mg.SO_4^{2-}.L^{-1})[a]	20–50			
Maximum airflow (mL.min^{-1})[b]	21.4	21.4	31.9	31.9
Air (mL.min^{-1})	20	10	30	30
Rate LO_2/L_{sewage}	0.005	0.003	0.005	0.005
Height of air introduction (m)[c]	3	3	3	1
Duration (days)	47	39	77	50

[a]Influent sulfate concentration adopted for airflow estimation;
[b]Maximum airflow estimated considering maximum influent sulfate concentration and stoichiometric equations (see equations 1 and 2);
[c]Distance from the bottom of the reactor.
Note: The HRT of phases 3 and 4 was reduced due to obstruction problems in the sewerage pipe.

carbon dioxide (CO_2) were monitored with a portable gas analyzer (LANDTEC® GEM_{TM} 5000). Cumulative biogas production was measured using a wet gas meter (Ritter® TG05).

Temperature, dissolved oxygen (DO), pH and oxidation reduction potential (ORP) (Ag/AgCl reference electrode) were monitored by inserting multiparameter probes (Hach HQ40D) at the settler compartment of the reactors.

Chemical organic matter (COD), sulfate (SO_4^{2-}), thiosulfate ($S_2O_3^{2-}$) and settleable solids (Sset) were measured according to Standard Methods for Examination of Water and Wastewater [26]. Sulfide (S^{2-}) was measured according to Plas et al. [27] and elemental sulfur (S^0) was measured by extraction with chloroform and analyzed with high-performance liquid chromatography (HPLC) using a PRP-1, reversed-phase, HPLC column (dimensions: 15 cm-L ×4.1 mm-ID), as described by Henshaw et al. [28].

The chemical elementary characterization of the scum layer naturally formed and maintained inside the three-phase separator of the microaerated UASB reactor was determined by X-ray fluorescence analysis using Phillips PW-2400 equipment, after the scum was dried at 40°C until complete humidity removal was achieved. The X-ray fluorescence spectrometer was used to determine the elemental composition of the sample by measuring the fluorescent X-ray emitted when it was excited by a X-ray source, thus sulfur extraction from the raw sample was not required. The X-ray analysis resulted in several peaks, enabling the identification of the elements present in a specific sample.

Sulfur mass balance

Sulfur mass balance was determined according to equations 4 to 7. Influent sulfur (Sinfluent) was exclusively attributed to dissolved fraction in the sewage (I): dissolved sulfide (S^{2-}-S) and sulfate (SO_4^{2-}-S). Sulfur fractions in the investigated UASB reactors were divided in liquid ($S_{effluent}$), gas (S_{biogas}) and solid (S_{sludge}) phases. Influent ($Q_{influent}$) and biogas flowrates (Q_{biogas}) were considered to determine sulfur mass in both liquid and gas phases. All sulfur fractions were determined in terms of $mg.L^{-1}$, except for H_2S measurements in the biogas, reported as ppm. Biogas concentration was converted to $g.L^{-1}$ by adopting the stoichiometric coefficient of H_2S in biogas (K_{H2S}) under room temperature (25°C) and pressure (0.91 atm) conditions. Sulfur fraction in the solid phase of the investigated UASB reactors was estimated as shown in equation 7, considering elemental sulfur concentration in the sludge and total solids content in sludge (TS_{sludge}).

$$\text{Sulfur fractions - raw sewage Liquid phase}\quad S_{influent} = (I_{SO_4^{2-}-S} + I_{S^{2-}-S})x\ Q_{Influent} \tag{4}$$

$$\text{Sulfur fractions}-\text{UASB reactors Liquid phase}\quad S_{effluent} = (E_{SO_4^{2-}-S} + E_{S^{2-}-S} + E_{S^0-S} + E_{S_2O_3^{2-}-S})x\ Q_{Influent} \tag{5}$$

$$\text{Gas phase}\quad S_{biogas} = B_{H_2S}\ x\ Q_{Biogas}\ x\ K_{H_2S} \tag{6}$$

$$\text{Solid phase}\quad S_{sludge} = \frac{S_{S^0}}{TS_{Sludge}} \tag{7}$$

In order to obtain an accurate mass balance, three 24 h composite samples of raw sewage were collected for sulfate determination. The results obtained from the composite samples is shown in Table 4.

Table 4. Sulfate concentration determined in three composite samples of raw sewage influent to the UASB reactors.

	Sample 1	Sample 2	Sample 3	Average
Sulfate ($mg.SO_4^{2-}.L^{-1}$)	45.6	56.5	52.2	51.4

Statistical analysis

The statistical analysis of the experimental data was performed by using Statistica 6.0 Software, considering the Mann-Whitney U Test (nonparametric) to compare the performance of the investigated UASB reactors throughout the monitoring period. The results obtained from the statistical tests consider a significance level of 0.05.

Results and discussion

Biogas desulfurization

The biogas composition of both anaerobic and microaerated UASB reactors is summarized in Table 5 and presented in Figures 2–5. The duration of all operational phases (P1, P2, P3, and P4) resulted in 213 operational days.

The air flow rate of 20 $mL.min^{-1}$ added at 3 m from the bottom of the reactor (R2-microaerobic), under conditions of 7 h of HRT, provided a considerable decrease in hydrogen sulfide concentrations in the biogas. The average hydrogen sulfide concentrations in biogas for R1-anaerobic and R2-microaerobic were 43 ± 21 and $1{,}719 \pm 477.3$ ppm_v of H_2S, respectively. Hydrogen sulfide concentrations observed for the R2-microaerated remained below 70 ppm_v through the operational period, representing a sulfide removal efficiency of 98%. During phase 1, a stable hydrogen sulfide

Table 5. Biogas composition of the anaerobic and microaerated UASB reactors.

	Operational phases	UASB Reactor	H_2S (ppm)	CH_4 (% $_{v/v}$)	O_2 (% $_{v/v}$)	N_2 (% $_{v/v}$)	CO_2 (%$_{v/v}$)
P1	Air flow: 20 mL.min^{-1} Position: at 3 m from the bottom of the reactor	R_1-anaerobic	1719.0 (±454.2)	72.2 (±7.6)	0.7 (±0.2)	21.8 (±9.7)	6.3 (±2.4)
		R_2-microaerated	43.0 (±21.4)	52.0 (±8.5)	3.0 (±1.9)	40.2 (±7.8)	4.3 (±0.8)
P2	Air flow: 10 mL.min^{-1}	R_1-anaerobic	1826.4 (±62.9)	66.4 (±4.6)	0.4 (±0.2)	27.6 (±4.6)	5.9 (±0.6)
	Position: at 3 m from the bottom of the reactor	R_2-microaerated	1172.1 (±369.4)	62.3 (±9.2)	0.4 (±1.1)	32.0 (±9.3)	4.5 (±0.9)
P3	Air flow: 30 mL.min^{-1}	R_1-anaerobic	1145.7 (±347.4)	64.5 (±5.6)	0.8 (±0.3)	29.6 (±6.2)	5.4 (±0.5)
	Position: at 3 m from the bottom of the reactor	R_2-microaerated	44.0 (±95.9)	47.5 (±10.7)	5.8 (±2.3)	42.6 (±9.7)	5.0 (±1.0)
P4	Air flow: 30 mL.min^{-1}	R_1-anaerobic	1079.0 (±320.1)	58.4 (±6,8)	0.6 (±0.2)	36.1 (±7.9)	4.9 (±0.8)
	Position: at 1 m from the bottom of the reactor	R_2-microaerated	52.0 (±79.4)	41.8 (±6,4)	7.0 (±1.8)	46.2 (±7.2)	4.6 (±1.1)

Median values. Standard deviations in parenthesis.

removal was observed due to the air introduction, resulting in a biogas upgrade in terms of sulfide concentration (Figure 2).

In the second phase of the study (P2), the air flow rate decrease from 20 to 10 mL.min^{-1} did not improve hydrogen sulfide removal. Both investigated reactors presented similar hydrogen sulfide concentrations during phase 2 and the presence of oxygen in the biogas was not observed, as shown in Figure 3. In light of the operational conditions adopted, it is possible to assume that the low air flow rate was a limiting factor for achieving satisfactory sulfide removal efficiencies during phase 2. Considering the microaeration apparatus used, it is possible that the low efficiencies observed may be due to difficulties in setting low air flow rates in the pilot-scale system.

Increasing the air flow rate in the subsequent phase (P3) from 10 to 30 mL.min^{-1} and reducing the HRT from 7 to 5 h resulted in unstable hydrogen sulfide concentrations in the biogas of the microaerated reactor. Nevertheless, the highest hydrogen sulfide concentration observed during this phase was 341 ppm_v, a value expressively lower than those observed in the biogas of the anaerobic reactor (see Figure 2).

Phase 4 consisted of supplying the same amount of air at a lower height (1 m from the bottom of the reactor), resulting in 324 ppm_v as the highest hydrogen sulfide concentration. The median sulfide removal efficiency from biogas was 96% and 94% for phases 3 and 4, respectively. According to the Mann-Whitney U Test, significant differences in hydrogen sulfide concentrations in biogas during phases 3 and 4 were not observed. Therefore, it is possible to assume that the height of the air supply did not affect hydrogen sulfide removal from biogas under the investigated conditions. That may be possibly due to excessive aeration since it was expected that air introduction at lower heights (1 m from the bottom of the reactor) would result in lower sulfide removal efficiencies due to possible oxygen uptake by different pathways along the digestion zone.

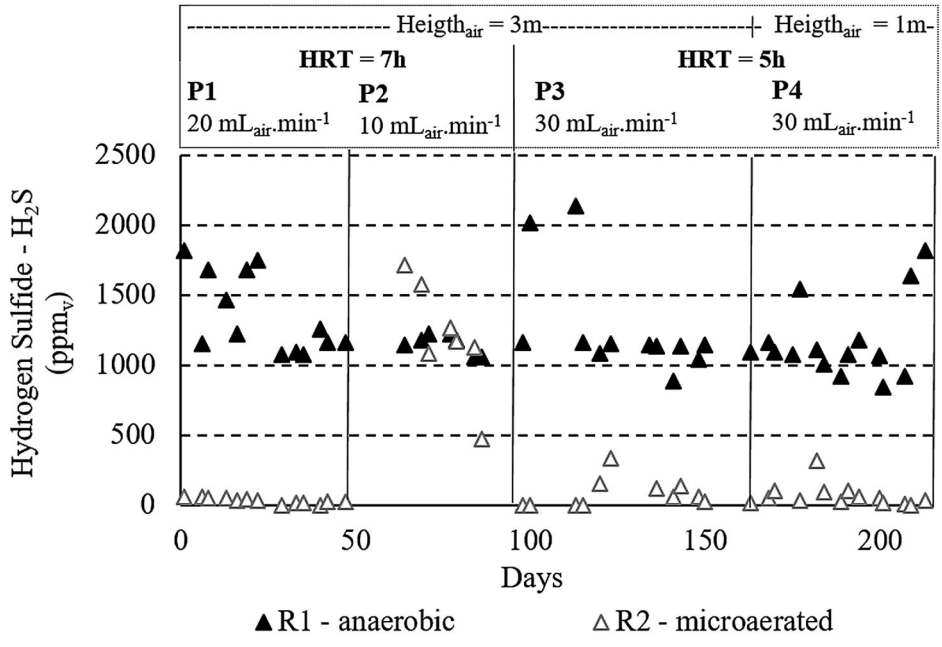


Figure 2. Hydrogen sulfide concentrations in the biogas of the anaerobic (R1-anaerobic) and microaerated (R2-microaerated) reactors under the tested conditions.

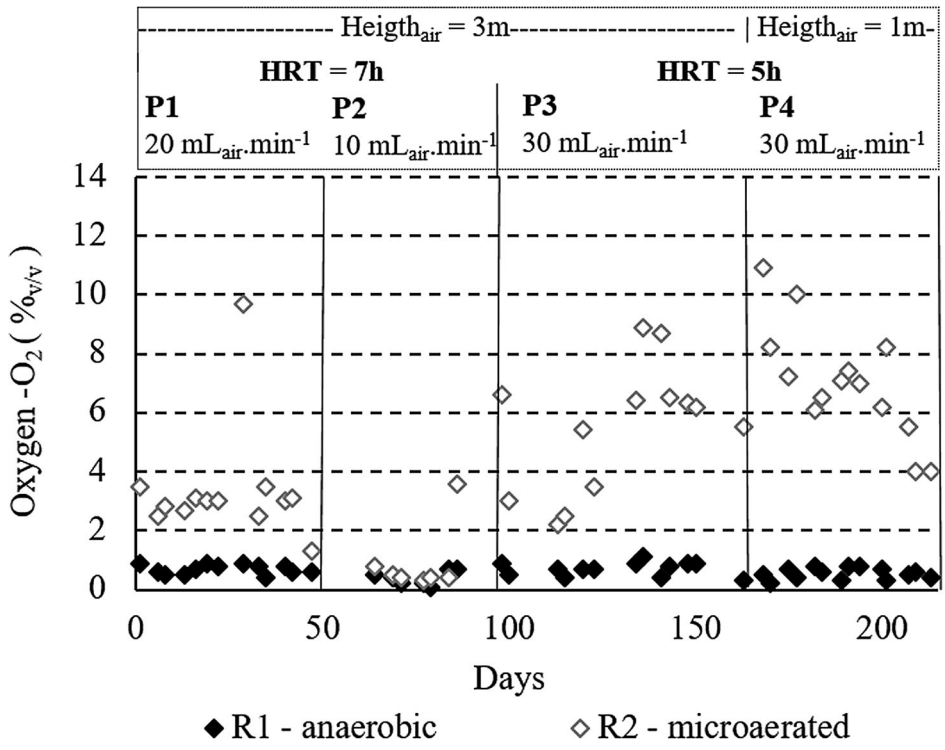


Figure 3. Oxygen concentrations in the biogas of the anaerobic (R1-anaerobic) and microaerated (R2-microaerated) reactors under the tested conditions.

In terms of oxygen concentrations in the biogas, greater concentrations were observed during phases 3 (5.2% ± 2.3) and 4 (7.0% ± 1.9) in comparison to phase 1 (3.4% ± 1.9). Such differences in oxygen concentrations may be due to the lower hydraulic retention time of 5 h adopted during phases 3 and 4. The lower HRT

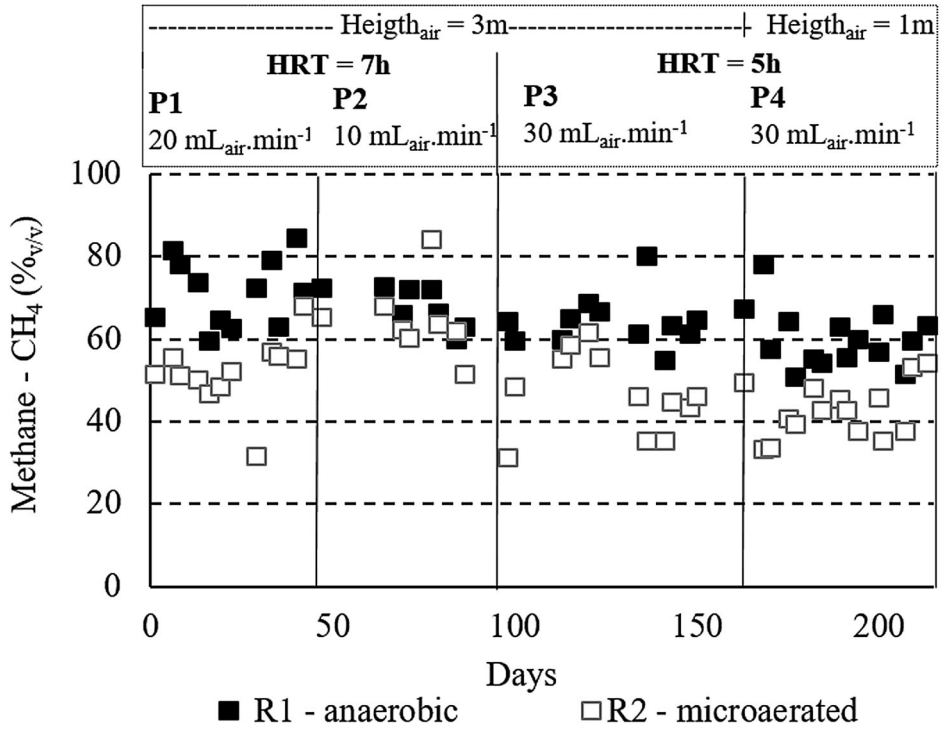


Figure 4. Methane concentrations in the biogas of the anaerobic (R1-anaerobic) and microaerated (R2-microaerated) reactors under the tested conditions.

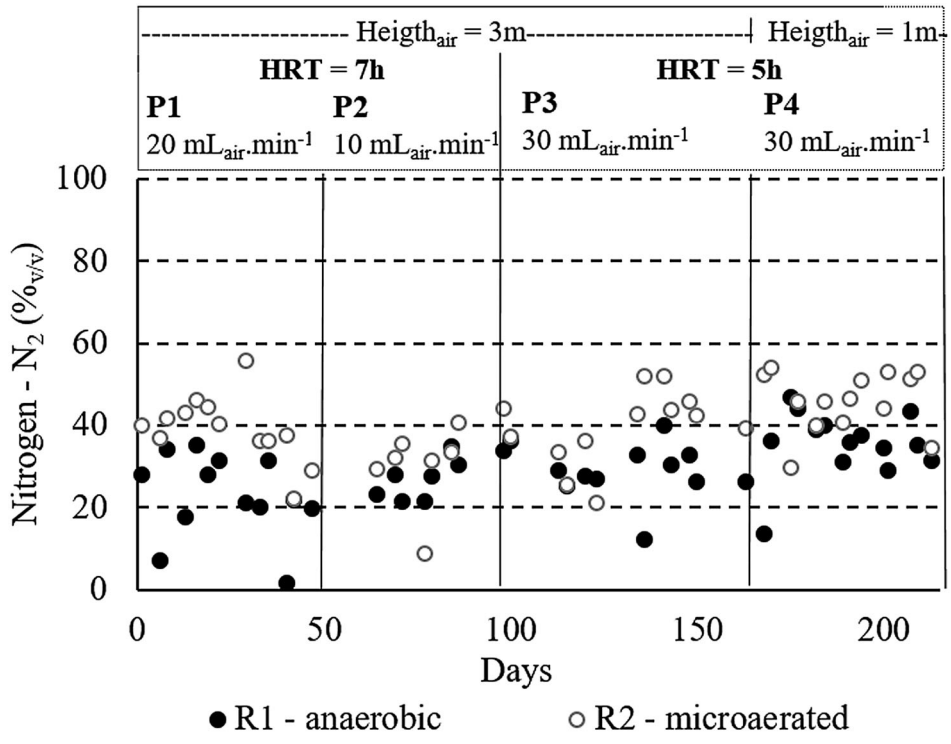


Figure 5. Nitrogen concentrations in the biogas of anaerobic (R1-anaerobic) and microaerated (R2-microaerated) reactors under the tested conditions.

associated with higher upflow air velocity might have favored air stripping into the biogas. It is possible to assume that the greater oxygen content in biogas may have affected oxygen uptake by facultative bacteria or sulfide oxidizing bacteria.

Regarding sulfide removal efficiencies from biogas by microaeration, Kobayashi et al. [29] demonstrated that the use of microaeration resulted in average removal efficiency of hydrogen sulfide from biogas of 68.2%. However, a wide range of hydrogen sulfide concentrations was observed, ranging from 0 to 2500 ppmv, probably due to the increment observed for biogas production (from 481 to 600 $m^3.d^{-1}$) and consequent greater H_2S production.

Jeníček et al. [11] investigated full-scale sludge digesters under microaerated conditions and verified that biogas desulfurization by microaeration is now a consolidated technology and the lack of operational expertise has been overcome. Although the start-up of microaerobic desulfurization may take a few weeks for biomass acclimatization and adjusting air flow rates, high hydrogen sulfide removal efficiencies may be achieved (99%) even under initial sulfide concentrations above 4,000 $mg.m^{-3}$ (corresponding to 2,876 ppm_v). Sousa et al. [23] also reported hydrogen sulfide removal above 90% for anaerobic synthetic sewage treatment with 0.28%$_{v/v}$ (corresponding to 2,800 ppm_v) of hydrogen sulfide content in biogas. In this context, this research contributes towards the use of microaeration as an alternative for biogas desulfurization from typical domestic sewage containing H_2S concentrations up to 2,135 ppm_v, aiming at demonstrating the feasibility of this technique for high hydrogen sulfide removal efficiencies under conditions similar to those present in full-scale systems.

Concerning methane content in biogas, it is possible to observe that the microaeration led to a decrease in methane concentration up to 20%, as shown in Figure 4. Greater reduction in methane concentration was observed during phase 1, from 72% to 52%. A similar reduction was observed in phases 3 and 4, of 17% (P3 - from 64 to 47% and P4 - 59 to 42%), probably due to the adoption of the same air ratio and sewage flow, of 0.005 LO_2/L_{sewage} in both phases. Although lower methane concentrations were determined due to the microaeration, the concentrations were above 42%, indicating thus that energy use would still be feasible.

Results obtained during phase 1 were not expected since the adoption of higher HRT (7 h) along with air and sewage flow ratio similar to the one considered during phases 3 and 4 (0.005 LO_2/L_{sewage}) should lead to higher methane concentrations than those obtained during the following phases. That was assumed from

the adoption of similar air/sewage flows under conditions of lower inflow rate, suggesting lower biogas dilution is due to the contribution of N_2 dissolved in the sewage stripped to the biogas [7].

Besides the nitrogen naturally dissolved in the sewage, biogas dilution in the present study may also be associated with atmospheric nitrogen introduction as a result of microaeration, as Figure 5 shows. The increase of nitrogen concentrations in the biogas of the microaerated reactor was 19% in the first phase (R1-anaerobic: 21%$_{v/v}$ and R2-microaerated: 40%$_{v/v}$), 14% in phase 3 (R1-anaerobic: 29%$_{v/v}$ and R2-microaerated: 43%$_{v/v}$), and 10% in phase 4 (R1-anaerobic: 36%$_{v/v}$ of N_2 and R2-microaerated: 46%$_{v/v}$ of N_2).

According to Kobayashi et al. [29], methane dilution in the biogas in microaerated systems is proportional to the amount of nitrogen introduced. Sousa et al. [23] presented a decrease of 14% of methane content in biogas due to microaeration. The authors concluded that biogas dilution may also be associated with oxygen uptake, sulfide removal, the residence time of air in the bulk liquid, etc. Biogas dilution by nitrogen is the main disadvantage of using air instead of pure oxygen for aeration purposes [29].

Regardless of the decrease in methane concentration in the biogas of the microaerated reactor, the energetic use of biogas would still be feasible since its application for energy cogeneration requires methane concentrations greater than 40 or 45%$_{v/v}$ to prevent operational problems [30].

Methane mass flow in the microaerated reactor during phases 3 and 4 was in the same range as that observed under control conditions (R1-anaerobic), as presented in Figure 6. Median values observed for methane mass flow of the microaerated reactor during phase 4 (air flow of 30 mL.min^{-1} introduced 1 m from the bottom of the reactor) were quite similar to the values observed for the control reactor, 36.7 and 37.0 g.d^{-1}, respectively. Therefore, it is possible to assume that methane mass flow is influenced by the presence of oxygen in the upper compartment of the digestion zone. Nevertheless, future research should address the effect of air inlet position on methane production to confirm the results obtained in phase 4.

Ramos and Fdz-Polanco [14] evaluated the effect of microaeration on the performance of an anaerobic sludge digester under overload conditions. The authors observed the improvement of biogas quality due to hydrogen sulfide removal. Greater methane production was observed under microaerated conditions (354 NmL.gVS^{-1}) as opposed to the anaerobic reactor (342 NmL.gVS^{-1}).

Regarding carbon dioxide concentrations in the biogas, low values were observed for both bioreactors under all tested conditions, within the range typically observed for biogas produced from sewage treatment (5 to 10%) [7], as shown in Figure 7. In addition, the dilution effect of air introduction on CO_2 concentration was not observed, with average values ranging from

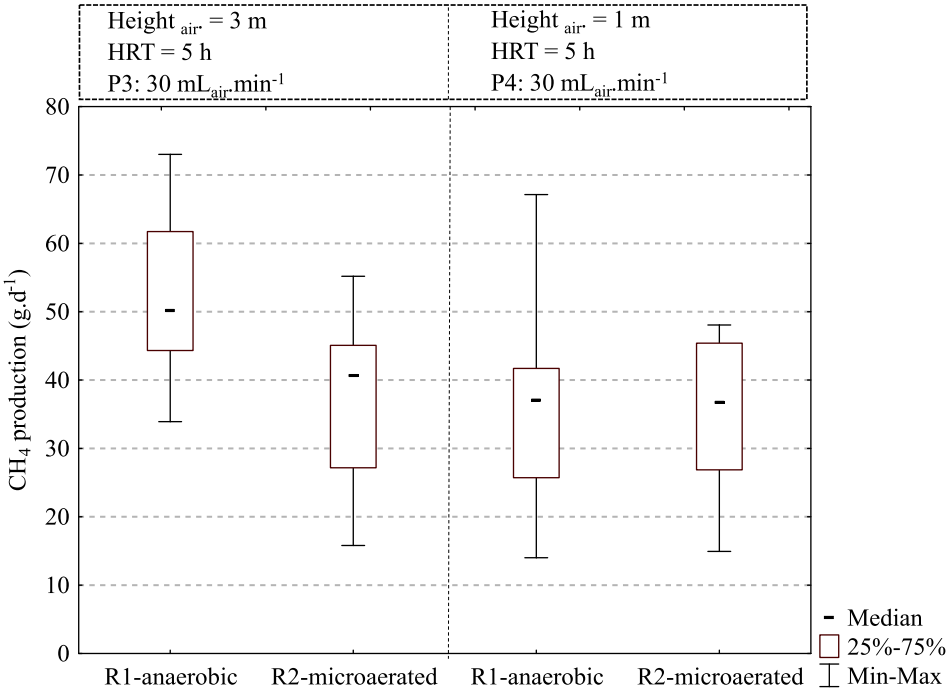


Figure 6. Methane production in the anaerobic and microaerated UASB reactors under distinct operational conditions. Note: methane production was not determined for all tested conditions due to equipment limitations.

4.5 to 6.8% in both reactors (R1-anaerobic and R2-microaerated). The CO_2 concentrations observed indicate that the greater amount of this compound remains in the effluent as bicarbonate ions. The CO_2 partial pressure is relatively low considering the alkalinity (200 to 250 mgCaCO3.L^{-1}) and pH (6.9 to 7.5) present in the anaerobic treatment of domestic sewage, therefore, a great portion of the CO_2 remains in the bulk liquid [31].

It is worth mentioning that high CO_2 concentration in biogas compromises its heating power and limits the economic feasibility of use [3]. Andriani et al. [3] emphasized that energetic use of biogas requires proper upgrade steps, including the removal of steam, H_2S, CO_2, siloxane, and trace compounds. However, Silveira et al. [32] reported that biogas is mainly used in sewage treatment plants for sludge drying (simplified thermal use with low purification required) and electrical/thermal energy cogeneration with gas engine operation (high complexity with greater purification required) [32].

Biogas composition required for energy generation may differ according to the equipment manufacturers. Biogas dilution by air addition observed in this study may affect and limit the potential of biogas use considering the equipment currently commercially available. However, in light of the advantages of H_2S removal from biogas by microaeration (i.e. simplicity and robustness), the development of new conversion systems considering lower methane content would improve biogas use, thus reducing energy losses and supporting low-cost techniques such as microaeration application in anaerobic reactors, without the need to add a further step for biogas upgrade [14].

Elemental sulfur formation

In order to investigate the formation of elemental sulfur in the scum layer maintained inside the three-phase separator, the chemical elementary characterization was performed by the X-ray fluorescence (FRX) analysis, as shown in Figure 8. The peaks identified correspond to average values for (S), silicon (Si), phosphorus (P), aluminum (Al), iron (Fe), and magnesium (Mg). Other elements were identified at low and trace levels. The chemical analysis is semiquantitative and only indicates the presence of the compound in the sample. Thus, it is preliminary evidence of elemental sulfur formation, specifically in the scum layer inside the three-phase separator, as a result of hydrogen sulfide oxidation due to air introduction in the bulk liquid.

Operational conditions of the investigated system

The operational conditions measured at the settler compartment of the investigated reactors and setteable

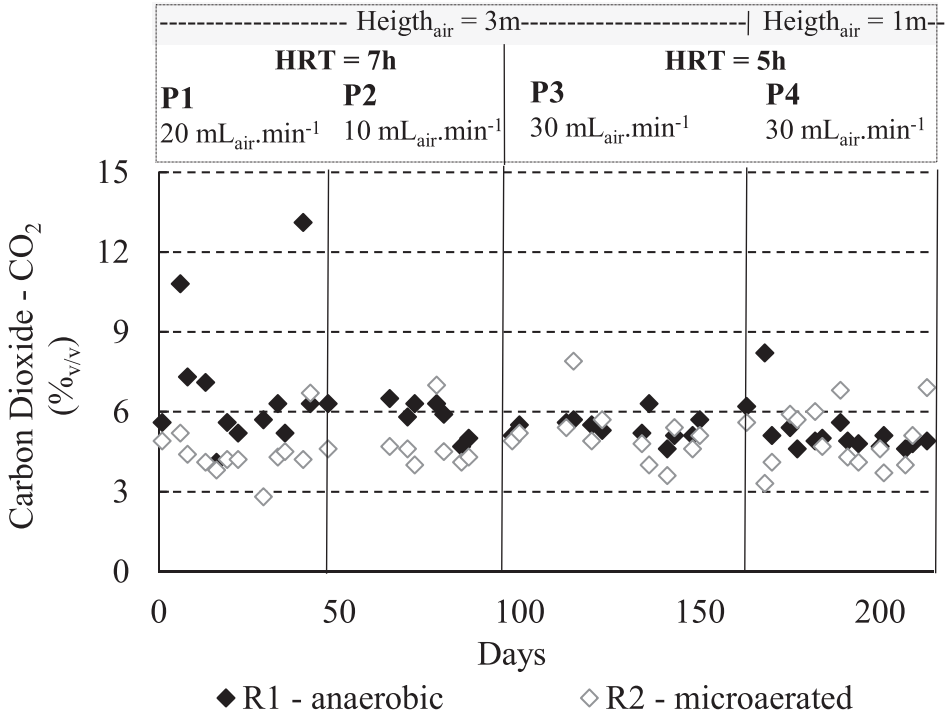


Figure 7. Concentrations of carbon dioxide in the biogas produced in the anaerobic and microaerated reactors throughout the experimental phases.

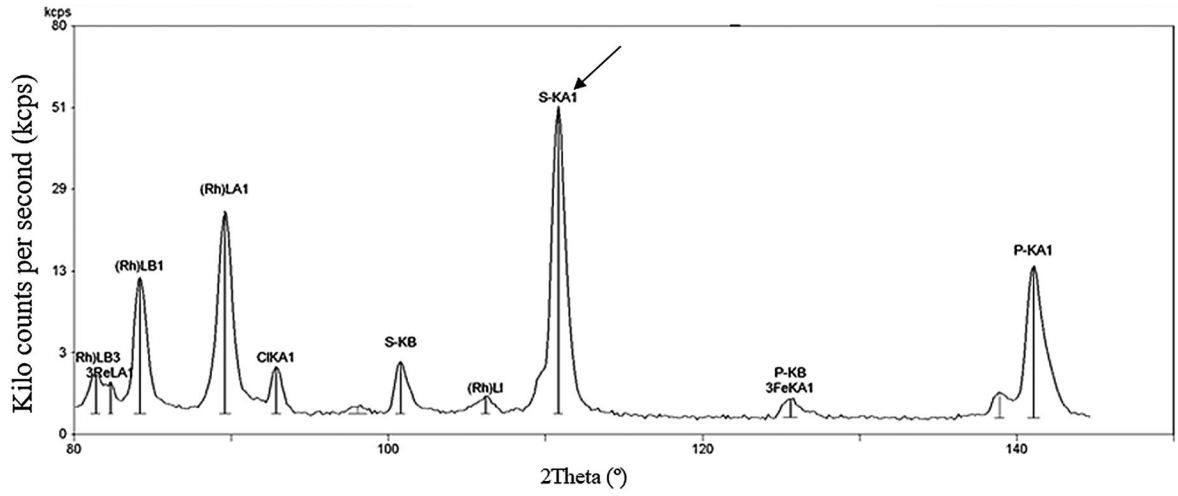


Figure 8. Chemical characterization of the scum layer inside the three-phase separator during phase 1 by X-ray fluorescence analysis.

solids in the effluent of both reactors are presented in Table 6. In general, the operational data of the anaerobic and microaerated UASB reactors were similar. The pH values remained close to neutrality, in the range commonly observed in sewage treatment, without significant variation between anaerobic and microaerated conditions (statistical tested with a significance level of 0.05). Mesophilic temperatures prevailed during the monitored period (21 to 24 °C).

Dissolved oxygen concentrations were equal or slightly superior at the settler compartment of the microaerated reactor. It is worth mentioning that previous studies have reported DO as an inaccurate parameter for microaeration conditions control since the amount of dissolved oxygen are usually lower than detection limit of conventional DO electrodes [14,17,33,34]. According to Jenicek et al. [34], the oxygen supplied in microaerated systems results in a limited increase in the ORP. Such effect was not observed, as the median ORP values observed for both UASB reactors ranged from -300 to -310 mV. Since the operational parameters were measured at the settler compartment of the UASB reactors and the top of the reactors was open to the atmosphere, it is possible to expect that this configuration may have affected the measurements due to external interferences.

Settleable solids concentrations in the effluent did not show significant differences under the investigated conditions. Therefore, one can expect that the additional turbulence due to the microaeration positioned at the digestion compartment did not promoted solids washout and loss in the effluent.

Organic matter and sulfide removal from the bulk liquid

The performance of the investigated systems in terms of organic matter removal is shown in Figure 9. Higher median COD removal efficiencies were observed for the microaerated reactor during phases 3 and 4 (Phase 3: R1-anaerobic: 64.3% and R2-microaerated: 68.8%; Phase 4: R1-anaerobic: 63.7% and R2-microaerated: 74.0%). However, the statistical analysis did not show any significant differences at a 5% significance level throughout the operational phases. The results presented in Figure 9 indicate that air addition in the microaerated reactor did not affect the reactor performance in terms of organic matter removal, as

Table 6. Operational conditions of the anaerobic and microaerated UASB reactors.

	Fases	UASB Reactor	pH	T (°C)	ORP (mV)	DO ($mg.L^{-1}$)	Sset[a] ($mL.L^{-1}$)
P1	Air flow: 20 $mL.min^{-1}$	R_1-anaerobic	6.9 (±0.2)	21.1 (±1.5)	-310.4 (±4.3)	0.22 (±0.03)	0.5 (±0.1)
	Position: at 3 m from the bottom of the reactor	R_2-microaerated	6.9 (±0.3)	20.9 (±1.6)	-309.5 (±4.2)	0.32 (±0.05)	0.5 (±0.4)
P3	Air flow: 30 $mL.min^{-1}$	R_1-anaerobic	6.9 (±0.2)	24.2 (±1.4)	-304.7 (±20.2)	0.20 (±0.08)	1.0 (±0.5)
	Position: at 3 m from the bottom of the reactor	R_2-microaerated	6.8 (±0.2)	23.9 (±1.5)	-305.4 (±14.7)	0.24 (±0.04)	0.9 (±1.1)
P4	Air flow: 30 $mL.min^{-1}$	R_1-anaerobic	6.9 (±0.3)	23.5 (±0.2)	-307.0 (±8.8)	0.16 (±0.03)	0.6 (±0.2)
	Position: at 1 m from the bottom of the reactor	R_2-microaerated	6.9 (±0.2)	23.5 (±1.2)	-303.5 (±12.4)	0.18 (±0.03)	0.5 (±0.4)

Median values; standard deviations in parenthesis;
[a]settleable solids.
Note: P2 was not included due to the absence of oxygen in the biogas.

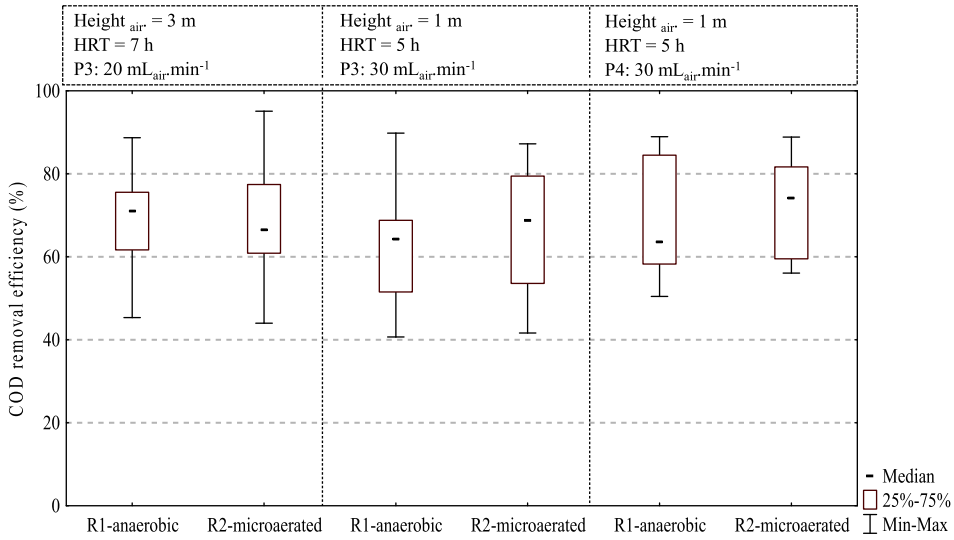


Figure 9. Organic matter removal efficiency in the anaerobic and microaerated UASB reactors.

corroborated by Sousa et al. [23] (0.0098 $LO_2/L_{influent}$), a previous study that has investigated the effect of microaeration on anaerobic reactors treating low-strength sulfate-containing synthetic wastewater.

Nevertheless, research in the anaerobic field has indicated better degradability of organic compounds and an increase of microbial diversity under microaerated conditions due to coupling anaerobic and aerobic conditions. Lower biomass yield of facultative bacteria was observed under anaerobic conditions and an increase in bacterial diversity was observed in the presence of oxygen [15]. Positive impacts include hydrolysis improvement and consequent growth of hydrogenotrophic methanogens, possibly favoring pH buffering under overload conditions, increase in biogas production and methane due to higher acetoclastic activity rates [15,16].

Regarding sulfur compounds in the bulk liquid, no significant differences were observed between the investigated reactors during the operational phases, as presented in Table 7. It was observed that a portion of the influent sulfur compounds (51.4 $mgSO_4^{2-}$.L and 0.5 $mg.S^{2-}$.L) was converted to sulfide, thiosulfate, and elemental sulfur in the UASB reactors, regardless of the operational condition (with or without microaeration). Therefore, it is possible to assume that oxygen supply in the microaerated reactor was insufficient or mass transfer limitations from the gas to the bulk liquid prevented dissolved sulfide oxidation. In this context, anaerobic conditions prevailed during the operational phases and the activity of sulfate-reducing bacteria remained unchanged in the microaerated reactor [10].

Conversely, Sousa et al. [24] observed an abatement of sulfide and sulfate in a UASB reactor treating domestic sewage equipped with an air supply above the digestion zone. The authors observed sulfide and sulfate concentrations in the UASB reactor effluent lower than typical concentrations of anaerobically treated sewage (S^{2-}: 0.5 $mgS.L^{-1}$ and SO_4^{2-}: 2.1 $mgS.L^{-1}$), indicating thus that only 5% of influent sulfur was reduced to sulfide. Regarding elemental sulfur formation, 4.6 $mgS.L^{-1}$ was quantified in the UASB reactor effluent, representing 46% of the influent sulfur load. Such results indicate that air introduction above the digestion zone favored partial sulfide oxidation to elemental sulfur.

Sousa et al. [24] investigated the occurrence of sulfur oxidized compounds in the effluent of a UASB reactor

Table 7. Sulfur compounds in the effluent of the anaerobic and microaerated UASB reactors.

	Phases	UASB Reactor	Sulfate SO_4^{2-}	Sulfide S^{2-}	Thiosulfate $S_2O_3^{2-}$ mgS.L^{-1}	Elemental sulfur S^0
P1	Air flow: 20 mL.min^{-1} Position: at 3 m from the bottom of the reactor	R1-anaerobic	10.2 (±4.1)	7.4(±1.4)	-	3.8 (±2.4)
		R2-microaerated	11.7 (±2.1)	7.5 (±1.4)	-	3.6 (±2.5)
P3	Air flow: 30 mL.min^{-1} Position: at 3 m from the bottom of the reactor	R1-anaerobic	8.9 (±2.4)	9.6 (±2.1)	10.7 (±2.7)	4.2 (±2.3)
		R2-microaerated	9.7 (±2.4)	8.8 (±2.8)	9.8 (±2.9)	3.2 (±1.9)
P4	Air flow: 30 mL.min^{-1} Position: at $_1$m from the bottom of the reactor	R1-anaerobic	8.2 (±2,0)	7.6 (±2.8)	7.8 (±1.4)	3.2 (±1.6)
		R2-microaerated	7.5 (±3,2)	8.4 (±2.8)	7.5 (±3.6)	5.0 (±1.4)

Median values. Standard deviations in parenthesis. *Inconclusive results or unmonitored parameter
Note: P2 was not included due to the absence of oxygen in the biogas.

applied to domestic sewage treatment and equipped with an aeration device above the digestion compartment, with HRT of 12 h and 0.12 LO_2/L_{sewage}. The authors observed sulfide and sulfate concentrations (0.5 and 2.1 $mgS.L^{-1}$, respectively) considerably lower than those observed in this study (as shown in Table 7). Sousa et al. [24] reported that only 5% of influent sulfur (approximately 10 $mgS.L^{-1}$) converted to sulfide, while 46% was recovered in the effluent as elemental sulfur (4.62 $mgS.L^{-1}$), thus indicating that the microaeration above the digestion compartment of the UASB reactor favored the partial oxidation of reduced sulfide species.

Although anaerobic conditions have prevailed in the digestion zone of the microaerated reactor investigated in this study, it was possible to identify the formation of thiosulfate and elemental sulfur in the effluent of both reactors. According to Janssen et al. [35], under limiting oxygen conditions (<0.1 $mg.L^{-1}$), elemental sulfur is the main sulfide biological oxidation final product. Previous

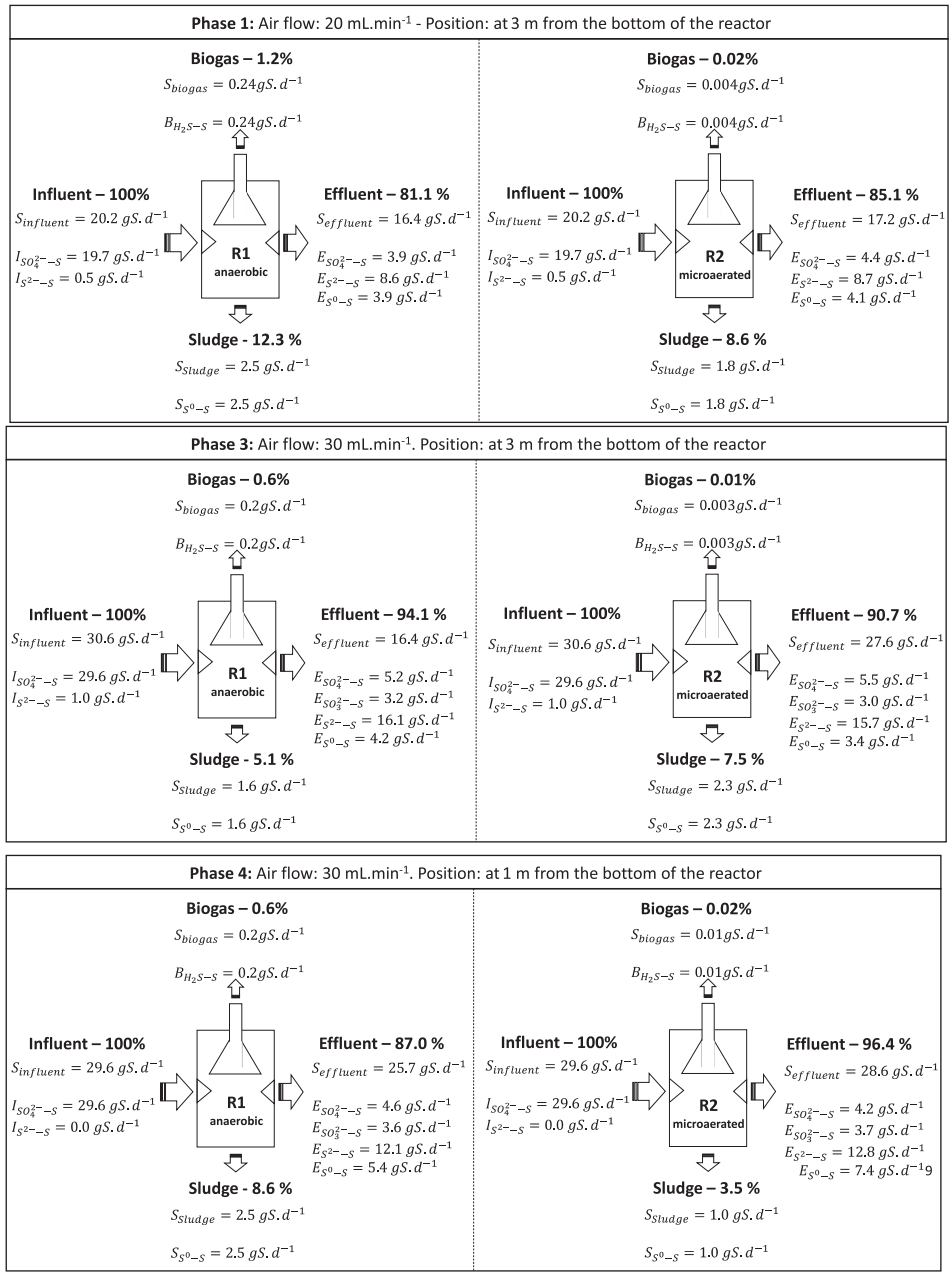


Figure 10. Sulfur mass balance in the anaerobic and microaerated reactors. Notes: Sinfluent, Seffluent, Sbiogas and Ssludge: sulfur load in the influent (raw sewage), effluent, biogas and sludge, respectively. Sulfur fractions in the effluent (E), biogas (B) and sludge (S) consist of sulfate (SO_4^{2-}), thiosulfate (SO_3^{2-}), sulfide (S^{2-}), hydrogen sulfide (H_2S) and elemental sulfur (S^0). P2 was not included due to the absence of oxygen in the biogas.

studies indicate that spontaneous thiosulfate formation is related to high sulfide loads in the presence of oxygen and under alkaline conditions, due to chemical oxidation [12,33,36]. It is possible to assume that thiosulfate and elemental sulfur present in the effluent of both UASB reactors were formed in the settler compartment due to oxygen incorporation from the atmosphere since the top of the UASB reactor is open (oxygen levels in the range of 0.1 and 0.2 $mg.L^{-1}$).

Cabral et al. [37] evaluated the performance of bioreactors post-treating anaerobic effluent exposed to sunlight with microorganisms naturally present in UASB reactor effluent under microaerated (airflow of 10 $mL.L_{influent}^{-1}.h^{-1}$ or 2 $mLO_2.L_{influent}^{-1}.h^{-1}$) and non-aerated conditions and HRT of 6 and 3 h. Although higher sulfide median removal efficiency from the anaerobic effluent was observed for the microaerated reactor (79%) when compared to the anaerobic bioreactor (61%) under HRT conditions of 6 h no statistically significant differences were observed. According to Cabral et al. [37], the sunlight enabled phototrophic microorganisms growth and promoted sulfide oxidation.

According to Cabral et al. [37], exposure to sunlight favored the growth of photosynthetic bacteria, allowing sulfide oxidation under anaerobic conditions. Sousa et al. [24] noticed an abatement of sulfide concentrations 19.0 to 0.9 $mgS.L^{-1}$ in the effluent of an anaerobic filter open to the atmosphere. The authors attributed sulfide removal to sunlight exposure and gas transfer within the anaerobic unit. Under these conditions, reduced sulfur compounds present in the bulk liquid of the anaerobic filter were oxidized to elemental sulfur and sulfate by photoautotrophic and/or photoheterotrophic bacteria.

In light of possible sulfide oxidation pathways from the anaerobic effluent under distinct operational conditions and configurations of treatment systems, further studies should address the influence of the settler compartment of UASB reactors open to the atmosphere on sulfur conversions. The improvement of the microaeration technique applied to low sulfate-containing sewage is important, aiming at achieving high sulfide removal from the gas phase with less methane dilution in the biogas.

The results obtained in this study demonstrated that the microaeration is a relevant pathway for hydrogen sulfide removal from the biogas. However, this technique did not promote dissolved sulfide removal in the investigated reactor. Therefore, coupling microaerated UASB reactors with additional processes and/or additional units can solve such limitation. Regarding additional processes, one possibility is the autotrophic denitrification conversion by nitrate dosing in the influent, achieving efficiencies up to 100% [38]. A simple low-cost promising alternative of compact unit following the microaerated reactor is the packed desorption chamber, with dissolved sulfide removal efficiencies up to 90% [39,40].

Sulfur mass balance

The distribution of sulfur fractions in the anaerobic and microaerated reactors during operational phases 1, 3 and 4 is shown in Figure 10. The sulfur mass balance highlights that over than 80% of influent sulfur load is lost within the reactor's effluent, mainly as dissolved sulfide (40–50%). Such results are corroborated by Souza et al. [5], that have also reported dissolved sulfide as the main content of the effluent of UASB reactors applied to sewage treatment.

In this study, the microaeration was not sufficient to provide dissolved sulfide to elemental sulfur. However, the fraction of sulfur converted to hydrogen sulfide in the biogas under anaerobic conditions (0.6–1.2%), was mostly eliminated in the microaerated in the reactor (0.01–0.02%), similar than values reported by Sousa et al. [23], 0.02%.

Conclusions

The results obtained in this study demonstrated microaeration as an efficient strategy for biogas desulfurization. The air addition in the digestion zone of a UASB reactor treating domestic sewage showed hydrogen sulfide removal efficiencies from biogas greater than 90%. The decrease in methane concentrations in the biogas was observed, possibly caused by the dilution due to atmospheric air addition. Even considering biogas dilution, the concentrations reported (40 to 50% of methane content in biogas), would not affect the energetic use of biogas. The chemical elementary characterization of the scum layer inside the three-phase separator of the microaerated reactor showed the presence of elemental sulfur. The microaeration did not affect the overall performance of the reactor in terms of organic matter removal. However, dissolved sulfide removal was not achieved under microaerated conditions. Further studies are necessary to improve the microaeration technique in anaerobic reactors and evaluate the pathways that lead to sulfide oxidation.

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Data availability statement

Data available on request from the authors.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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